A Study of Meta-Cognitive Skills of Higher Secondary Students in Relatin to their Problem-Solving Ability and Critical Thinking

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Introduction:

- In the rapidly evolving landscape of education and the workplace, the ability to think independently, adapt to complex situations, and generate innovative solutions independently, has become increasingly-essential. The core Cognitive competencies are metacognition

- Critical thinking and problem-solving ability success in both academic and real-world context. These abilities are not isolated mental processes; rather, they are deeply influenced by one's capacity to reflect on, control, and adapt their thinking processes, a capability knownas meta-cognition.

Introduction continued:

- Meta-cognitive skills refer to the higher order thinking processes that involve active control over the Cognitive strategies used in learning and problem-solving, This includes immediate planning, monitoring and understanding one's performance and to be aware of their thought processes, recognize when they do something and adjust their strategies accordingly.

Rationale of the Study–

- The transition to higher secondary education marks critical juncture where student shift from rote memorization to complex, abstract thinking. At this stage, academic success depends heavily on student's ability to navigate unstructured problems and think critically, while traditional curriculum focus on what to learn, there is an urgent need to investigate how students learn.

Rationale of the Study continued:

- Understanding the interplay between metacognitive skills, problem solving and critical thinking is essential for bridging's the academic achievement gap and addressing the dynamic nature of 21st century skills.
- Studies consistently shows a strong, positive, and bidirectional correlation between meta-cognition skills, problem-solving abilities, and critical thinking . Meta-cognition, or "thinking about thinking," serves as a foundational process that enhances and regulates both problem-solving and critical thinking skills.

Statement of the Problem :

- A Study on Meta-cognition Skills of Higher Secondary Students in Relation to Their Problem-Solving Ability And Critical thinking.

- **Operational definitions of key terms:**

Meta-cognition skills: Students having Meta-cognition skills means that they are able to recognize their own cognitive abilities,direct their own training,evaluate their own learning,evaluate their performances,understanding what caused their successes or failure, and learn new strategies.

Statement of the Problem continued

- **Problem solving ability**: It is a systematic process that involves defining the problem, investigating its root cause, generating and evaluating potential solutions, and implementing and verifying the chosen solution.

- **Critical Thinking:** It is the process of analyzing available facts, evidence, observations and arguments to make sound conclusions or informed choices.

Review of related literature :

- A Literature review is a comprehensive survey and critical analysis of previously published scholarly works on a specific topic.

  Its primary purpose is to summarize existing knowledge, synthesize varying perspective, and identify research gaps to contextualize, and identify research gap to justify our research.

Indian Review

(i) **Rekha Dhyani (2018)** studied on the Meta-cognitive Awareness of Primary School Students. The study shows meta-cognitive ability of students do not affect by gender and locality. So, we needed innovative teaching-learning strategies in our primary school classrooms.These practices develop meta-cognitive ability of students in early school stage.This helps him in their future as a good planner, thinker, and organized person.

Indian Review continued

(ii) Verma & Gupta (2024) explored the impact of self-regulated learning on senior secondary students' meta cognitive abilities. Data were collected using the Meta cognition Inventory (MCI) developed by Punita Govil (2011) and the Self-Regulated Learning Scale by Gupta and Mehtani (2017). No significant variation was found in the meta cognitive abilities of students across different self-regulated learning levels. Similarly, no significant differences were found based on school type (private or public), gender, or the interaction between self-regulated learning and school type. The study also found no significant combined influence of gender, school type, and self-regulated learning on meta cognitive abilities.

Indian Review continued

(iii) Sindhwani & Rakhi (2019) investigated the impact of meta cognitive knowledge on senior secondary school students' academic performance in Haryana. A descriptive survey was conducted on 600 students selected through cluster random sampling. Data were collected using a Meta cognitive Knowledge Scale and an Achievement Test in English. The results showed that students with high meta cognitive knowledge performed better academically than those with average and low meta cognitive knowledge, while students with average meta cognitive knowledge also outperformed those with low meta cognitive knowledge.

Indian Review continued

(iv) Rekha (2010) explored the correlation of meta cognition from an educational perspective. The population consisted of class eleven students from CBSE English medium schools within 15-17 years. Out of which, six schools were selected through a simple random technique. The data was collected using Meta cognitive Awareness Inventory, Academic Self Concept Scale, Index of Learning style, Styles of learning and thinking, and a personal data sheet. A significant and strong relationship between academic achievement and meta cognitive awareness was found in the study.

Review of related literature : International Review

(i) Catador (2024) conducted a study to identify the correlation and determinants of students' success in mathematics using eight meta cognition domains—declarative, procedural, conditional, planning, information management strategy, comprehension monitoring, debugging strategy, and evaluation. This study used a descriptive-correlational research design. A sample of 272 senior high school students in Grade 11 was selected. A multiple-choice test toevaluate mathematical competence and the Meta cognitive Awareness Inventory to measure meta cognitive awareness were used to gather the data.

- Metacognitive awareness and mathematics performance were found to be significantly correlated, suggesting that high-achieving pupils successfully applied and controlled their metacognitive abilities.

  The study discovered that 75.3% of metacognitive domains contributed to students achievement and that higher awareness of metacognitive thinking was associated with improved mathematical ability. However, the only strategies that could predict students' success in mathematics were declarative, procedural, conditional, and debugging.

International Review continued

(ii) **Kante & Vani (2024)** aimed to assess and compare metacognitive awareness between genders (girls and boys) .

They comparedthem across seightsub-components:

(i) declarative knowledge
(ii) procedural knowledge
(iii) conditional knowledge (iv) planning
(v) information management strategies
(vi) comprehension monitoring
(vii) debugging strategies and
(viii) evaluation.

A sample of 100 late teenagers, all of whom were 18 years old, participated in the study, and it was found that they possessed metacognitive awareness. No significant difference was found between boys and girls (demonstrating good knowledge) across all sub-components, except in the subcomponent of comprehension monitoring, where girls outperformed the boys.

International Review continued

(iii) Aisyi et al. (2021) conducted a study focusing on meta cognitive aspects among students of tenth grade. The research indicated that the three types of meta cognitive knowledge—procedural, declarative, and conditional—are closely related to the learning process of the students, with procedural knowledge being the highest level, followed by declarative and conditional knowledge. This hierarchy suggested that students' ability to execute learning strategies effectively remained foundational to their overall understanding and application of knowledge.

Delimitation of the study

(i) This study will be confined to meta-Cognitive skills, problem solving ability and Critical thinking of Higher secondary school students.

(ii) This study will be confined to only higher secondary school students spanning from 15-18 years.

(iii) The study will be confined to the students studying in schools affiliated to CBSE.

(iv) This study will be confined to schools situated in Raipur district

Description of Variables.

- Variables are fundamental components of research that allow for the measurement and analysis of data. They can be defined as characteristic of properties that can take on different values. In research design understanding the type of variables and their roles facilitate the development of hypotheses, choice of methods, and the interpretation of results

Description of Variables continued

- Variables can be categorized based on their role in the study (such as independent and dependent variables), the type of data they represent (quantitative or categorical), there relationship to other variables. Understanding what constitute a variable and the other variables types available is a critical step in designing meaningful research.
- **DependentVariable**-Inthepresentresearchwork,thedependentvariablesareProblem Solving ability and Critical Thinking.
- **IndependentVariable**-Inthepresentresearchwork,Meta-cognitionSkillsis Independent Variable.

Objectives of the study.

Research objectives describe what your research is trying to achieve and explain why you are pursuing it. They summarize the approach and purpose of your study and help us to focus on our research.

(**i**) To study the meta-cognition Skills among higher secondary school students.

(**ii**) To study the Problem-Solving ability of Higher Secondary school student.

Objectives of the study continued

(**iii**) To study the Critical Thinking ability of Higher Secondary school students.

(**iv**) To study the relationship between meta-cognition skills and Problem-Solving ability among higher secondary school students.

(**v**) To study relationship between meta-cognition skill and Critical Thinking among higher secondary school students.

Hypothesis of the study

A research hypothesis is a testable, educated prediction about the relationship between two or more variables. It translates a broad research question into a specific measurable claim that can be supported or refuted by data.

(i) There will be significant relationship between the meta-cognition skills and problem-solving ability of higher secondary school student.

(ii) There will be significant relationship between the meta-cognition skills and Critical Thinking of Higher Secondary School students.

(iii) There will be significant relationship between the problem-solving ability and Critical Thinking of Higher Secondary School Student.

Hypothesis of the study continued

(iv) There will be significant relationship between the Meta-cognition skills and Problem-Solving ability of male student's higher secondary school.

(v) There will be significant relationship between the Meta-cognition skills and Problem-Solving ability of female student's higher secondary school.

(vi) There will be significant relationship between the Meta-cognition skills and critical thinking of among male student's higher secondary school.

(vii) There will be significant relationship between the Meta-cognition skills and critical thinking of among female student's higher secondary school.

(viii) There will be significant relationship between the Meta-cognition skill sandcritical Thinking of female Student of Higher Secondary Schools.

Hypothesis of the study continued

(ix) There will be significant relationship between the Meta-cognition skills and critical Thinking of female Student of Higher Secondary Schools.

(x) There will be significant difference between the Meta-cognition skills of Maleand female students of higher secondary School.

(xi) There will be significant difference between the Problem-solving ability of male and female students of Higher secondary school.

(xii) There will be significant difference between the critical thinking of male and female Students of Higher secondary school.

(xiii) Meta-cognition skills predicts problem solving ability.

(xiv) Meta-cognition skills predicts Critical Thinking ability.

Research Design

A research Design is a procedural plan that is adopted by a researcher to answer questions in a valid way. It is a logical and systematic plan prepared for directing a research study. In the present study, two types of research design will be considered, They are

(i) Co-relational design

(ii) Comparative research design

- **Research Methodology.**

Research methodology is the systematic approach researchers use to conduct studies. It includes various types like qualitative, quantitative and mixed methods. The study employs a quantitative research approach (Descriptive survey method).

Research Design continued

- **Population of the study**

- The population of the present study will be the students of higher Secondary schools located at Raipur district and affiliated with CBSE.

- **Sample of the study**

- The sample of the study will be the 400 students of higher secondary school.

- **Sampling Method**

- Researcher will use Simple Random Sampling. 400 Students will be selected from CBSE affiliated Higher Secondary schools. Out of which 200 male students and 200 female students will be selected randomly.

Description of Tools

For the present research, the researcher will use the following standardised tools for collecting the data.

- **Meta-cognition skills**: To measure this variable, the Meta-cognition skills cale developed by Madhu Gupta and Suman will be used.
- **Critical Thinking**: to measure this variable, the scale developed by Hemant, Lata Sharma and Priyamvada will be used.
- **Problem solving ability**: To measure this variable, problem solving ability test developed by L.N.Dubey and C. P. Mathur will be used.

Source of Data

- There searcher will collect data by primary sources.

Data Analysis

- The descriptive statistical techniques will be used to analyze data. Pearson-product-moment Correlation, regression and the inferential statistics like t-test and ANOVA will be used during the process of data analysis

Expected Outcome of present study

(i) Significant Positive Correlation: The study is expected to find a strong, positive, and significant correlation between individuals' meta-cognition skills and their problem-solving and critical thinking abilities. This means that individuals who demonstrate higher levels of meta-cognition awareness and regulation also tend to exhibit better skills in complex problem-solving and critical analysis.

(ii) Meta-cognition as a Predictor: Meta-cognition skills are likely to be identified as a significant predictor of performance in higher-order thinking tasks (problem-solving and critical thinking).

Expected Outcome of present study continued

(iii) Improved Performance with Intervention: If the study involves a training intervention, the experimental group receiving explicit meta-cognition instruction would be expected to show significant improvements in their problem-solving and critical thinking test scores compared to a control group receiving conventional instruction.

- The study of meta-cognition skills in relation to problem solving and critical thinking carries significant educational implications, highlighting the need to shift from traditional instruction methods to approaches that explicitly foster students' awareness and regulation of their own thinking processes.

Educational Implications of the study

- **Explicit Instruction of Meta-cognition Strategies**: Educators should directly teach specific meta-cognition strategies (planning, monitoring, and evaluation) rather than assuming students will acquire them automatically. This can be embedded within the curriculum content, not just taught as a standalone "thinking skills" lesson.
- **Modeling by Teachers**: Teachers should model their own thinking processes aloud while solving problems or engaging in critical thinking tasks. This "think-aloud" method makes the expert thought process visible to students helping them understand.
- **Self-Questioning**: Encouraging students to ask themselves questions like, "What is the problem asking?“ "What strategies could I use?" or "Does this make sense?" during tasks.

Educational Implications of the study continued

- **Problem-Based and Collaborative Learning**: Implementing teaching methods like Problem-Based Learning (PBL)challenges students to solve real-world problems and work in groups, which naturally promotes meta-cognition reflection, discussion of strategies, and peer feedback.
- **Focus on the Process, Not Just the Outcome**: Feedback and assessment should focus not only on the correctness of answers but also on the reasoning process and the strategies used. Error analysis activities, where students identify patterns in their mistakes and plan how to avoid them, are particularly valuable.
- **Fostering a Growth Mindset**: Cultivating a classroom culture where challenges and mistakes are viewed as opportunities for growth encourages students to take intellectualrisks and adapt theirstrategies, akeycomponent of meta-cognition and resilience.

Educational Implications of the study continued

- **Teacher Training**: Providing teachers with professional development and training courses on meta-cognitive skills is crucial to ensure they understand the theory and are equipped with effective instructional techniques to implement these practices consistently. By integrating these strategies, educational institutions can foster environments that not only improve academic performance but also develop self-directed, adaptable, and effective lifelong learners equipped with essential 21st-century problem-solving and critical thinking skills.
- **Modeling by Teachers**: Teachers should model their own thinking processes aloud while solving problems or engaging in critical thinking tasks. This "think-aloud" method makes the expert thought process visible to students, helping them understand how to plan, monitor, and evaluate effectively.
- **Integration of Reflective Activities**: Regular opportunities for self-reflection are essential. **Strategies include**: Reflective Journaling: Students can write about their learning experiences, challenges, and the strategies they used.

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Thank you !

Thank you for your kind attention.

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